Best Coding Practices (Revisited)

DATA201/422 Data Wrangling

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Today’s plan

- What are AI coding agents?
- Summary of today’s data-wrangling problem
- Work with an AI coding agent to solve the problem

AI coding agents

What are AI coding agents?

- Examples: Codex, Claude Code, Cursor
- Have access to the whole project directory
  - Read and write
- Take multi-step actions
  - For example, plan a code change, make the change, test, revise, and test again
- Often distinguish between “planning” mode and “agent” mode
  - Distinction especially strong in Cursor

Comparison with an AI chatbot

Chatbot	AI coding agent
Responds with text	Takes actions
Uses supplied context	Explores the project
Suggests code	Edits files
Describes commands	Runs commands
Proposes a solution	Proposes, tests, and revises a solution

Human-in-the-loop coding

- Responsibilities
  - Human: Design, review, accountability
  - AI agent: Detailed implementation, including writing code and testing
- Recommended workflow
  1. Collaborate on overall design
  2. Build step by step
- You are the boss, not the agent!

Today’s data wrangling example

Output

Internal in-migration, in England and Wales, in 2024

region	sex	age	inflow	population
Southeast	Female	30-34	3302	42010
etc	*etc*	*etc*	*etc*	*etc*

- CSV file with data on in-migration into each region
- Variables
  - *region* identifies a region in England or Wales (England has 9 regions, and Wales is one region)
  - *sex* is Female or Male, as reported by the UK statistics agency
  - *age* is age on 30 June 2024, using 5-year age groups up to 90+
  - *inflow* is the number of moves into the region from elsewhere in England and Wales between 1 July 2023 and 30 June 2024
  - *population* is the estimated number of residents on 30 June 2024

Inputs

- Data from the Office for National Statistics, in the United Kingdom
- Contents
  - Data on internal migration for England and Wales, by **region**, sex, and 5-year age group
  - Data on population in England and Wales, by **local authority**, sex, and 1-year age group
  - A lookup table matching each local authority to a region